

Chikungunya transmission and population susceptibility in Ceará, Brazil

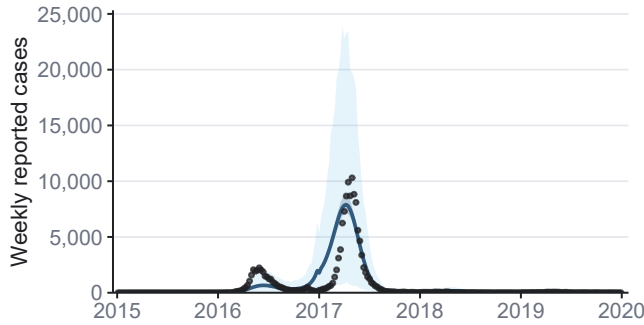
v4.6 fixed-q sweep, $q = 0.10$ (HMC PASS) | weekly observations, 2015-2019

A

Reported cases and posterior prediction

Shading: 95% predictive interval; dark band: 50% CrI

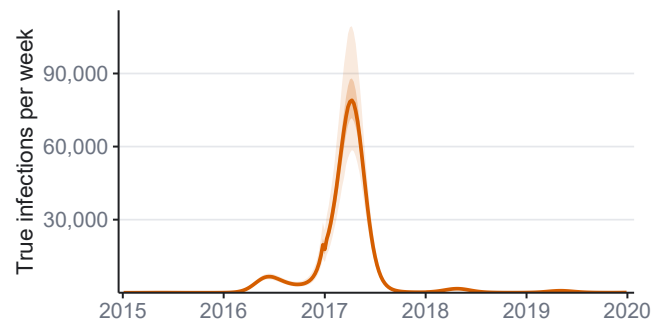
— Model expectation • Observed cases



B

Latent infection incidence

Median, 50% and 95% credible intervals

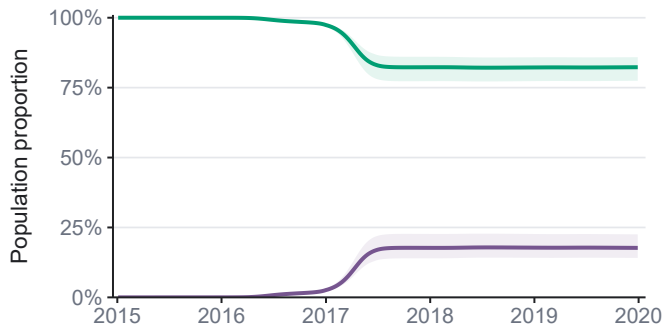


C

Population susceptibility and immunity

Closed population; lifelong immunity

— Infection-derived immune — Susceptible

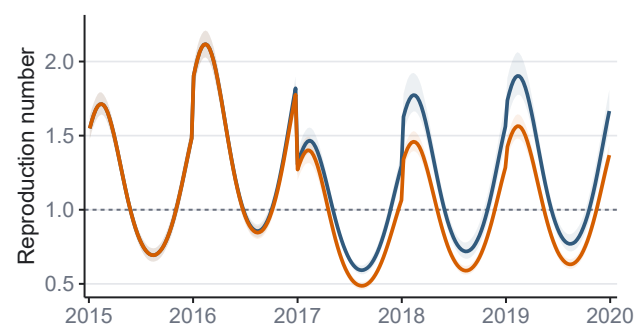


D

Time-varying reproduction numbers

$$R_{\text{eff}}(t) = R_0(t) \times S(t-1)/N$$

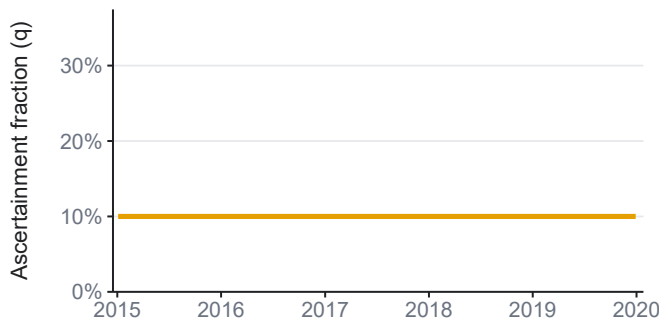
— $R_0(t)$ — $R_{\text{eff}}(t)$



E

Case ascertainment

q FIXED at 0.10 (not estimated -- v4.6 fixed-q sweep)



F

Cumulative infection and serological anchor

Yellow field: June-December 2018 survey; no site-level hierarchy

— Ceara immune fraction — Model Ceara immune fraction
— Observed Juazeiro seroprevalence

